

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

**20MEM102 – INTRODUCTION TO SELF-DRIVING CARS**

Duration: 3 Hours

Max. Marks: 100

**Note:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

| Unit – I   |                                                                                                                                                    | Marks | BT* | CO* | PO* |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.         | a) What is perception? Why perception is also a difficult problem in autonomous driving?                                                           | 10    | L*1 | 1   | 1   |
|            | b) Explain Operational Design Domain or ODD and discuss levels of automation.                                                                      | 10    | L2  | 2   | 1   |
| 2.         | a) How to select a hardware configuration by doing sensor coverage analysis both for longitudinal and lateral cases for highway and urban driving. | 10    | L1  | 1   | 1   |
|            | b) Explain the proprioceptive sensors. Also draw a sketch to emphasize the perception of sensors mounted on a car.                                 | 10    | L2  | 2   | 1   |
| 3.         | a) Explain the NHTSA's 12-part safety recommendations for autonomous driving.                                                                      | 10    | L2  | 1   | 1   |
|            | b) What are the most common sources of autonomous vehicle hazards? Explain.                                                                        | 10    | L2  | 2   | 1   |
| Unit – II  |                                                                                                                                                    |       |     |     |     |
| 4.         | a) Explain the safety ideology followed by General Motors.                                                                                         | 10    | L2  | 3   | 1   |
|            | b) Explain Coupled Lateral & Longitudinal dynamics that govern a vehicle using block diagram?                                                      | 5     | L2  | 4   | 1   |
|            | c) Explain steering system model for autonomous vehicles.                                                                                          | 5     | L2  | 4   | 1   |
| 5.         | a) Explain HAZOP and SOTIF.                                                                                                                        | 10    | L2  | 3   | 1   |
|            | b) Why dynamic model is important for self driving Vehicle? Formulate dynamic model for a rotational system with an example.                       | 10    | L1  | 4   | 2   |
| 6.         | a) Explain the five level safety approach designed by Waymo.                                                                                       | 10    | L2  | 3   | 1   |
|            | b) Describe typical powertrain model of a car with a block diagram. Write the dynamic equations of each powertrain components.                     | 10    | L1  | 4   | 2   |
| Unit – III |                                                                                                                                                    |       |     |     |     |
| 7.         | a) What is longitudinal cruise control? Explain with a block diagram.                                                                              | 10    | L1  | 5   | 1   |
|            | b) What is a controller? What are the commonly used controllers/compensators in control systems?                                                   | 10    | L1  | 5   | 1   |
| 8.         | a) What is a PID control? Explain.                                                                                                                 | 10    | L2  | 5   | 1   |
|            | b) Explain the architecture of vehicle control strategy with a block diagram.                                                                      | 10    | L2  | 5   | 2   |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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